

Earthquake Analysis Project

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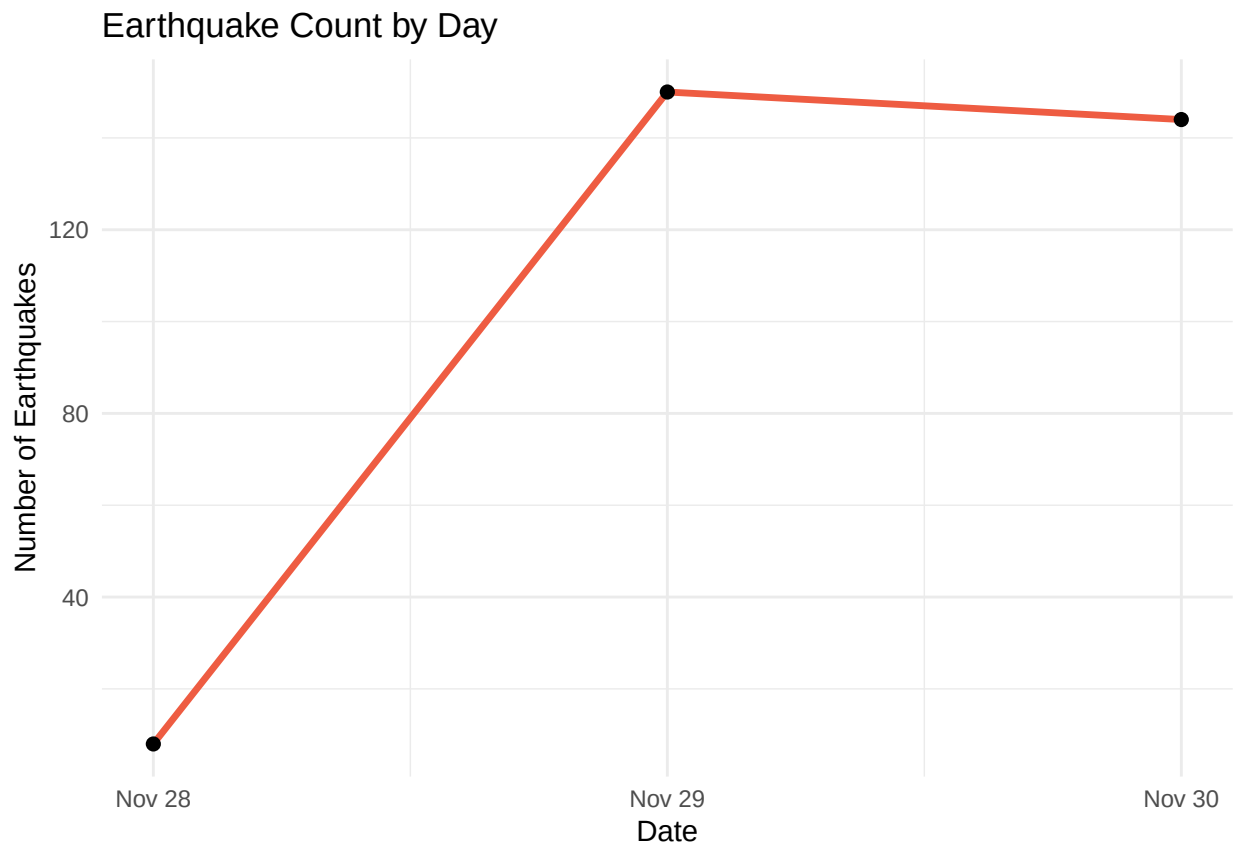
2025-11-30

```
df <- readr::read_delim("event_catalog_fixed.csv", delim = ";")

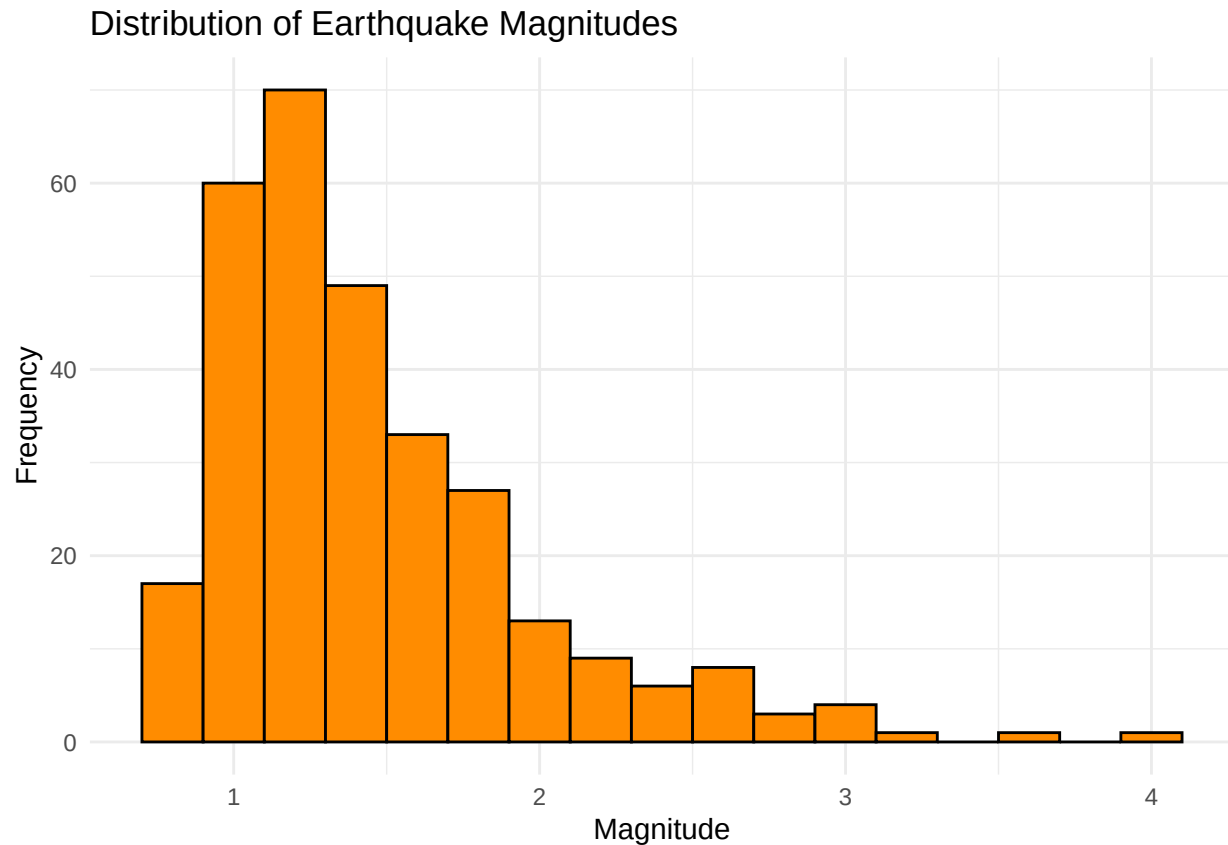
## Rows: 302 Columns: 9
## -- Column specification -----
## Delimiter: ";"
## chr (3): Date, Type, Location
## dbl (6): Longitude, Latitude, Depth, Rms, Magnitude, EventID
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
df <- df |>
  mutate(
    Date = dmy_hm(Date),
    Day = date(Date),
    Hour = hour(Date)
  )
head(df)

## # A tibble: 6 x 11
##   Date                Longitude Latitude Depth   Rms Type Magnitude Location
##   <dtm>                <dbl>    <dbl> <dbl> <dbl> <chr>    <dbl> <chr>
## 1 2025-11-30 22:23:00    33.3     39.1   7    0.51 ML        2.4 Kulu (Kony~
## 2 2025-11-30 22:22:00    36.7     38.2   7    0.82 ML        1.3 Gökşun (Ka~
## 3 2025-11-30 22:17:00    38.2     38.3  11.6   0.35 ML        2.2 Yeşilyurt ~
## 4 2025-11-30 22:16:00    28.2     39.2  11.0   0.48 ML        1.2 Sındırgı (~
## 5 2025-11-30 22:11:00    26.0     39.1   6.69   0.62 ML        1.8 Ege Denizi~
## 6 2025-11-30 22:05:00    28.3     39.2   5.88   0.51 ML        1.2 Sındırgı (~
## # i 3 more variables: EventID <dbl>, Day <date>, Hour <int>

df %>%
  filter(!is.na(Day)) %>%
  count(Day) %>%
  ggplot(aes(x = Day, y = n)) +
  geom_line(linewidth = 1.2, color = "tomato2") +
  geom_point(color = "black", size = 2) +
  labs(
    title = "Earthquake Count by Day",
    x = "Date",
    y = "Number of Earthquakes"
  ) +
  theme_minimal()
```



```
df %>%  
  ggplot(aes(x = Magnitude)) +  
  geom_histogram(binwidth = 0.2, fill = "darkorange", color = "black") +  
  labs(  
    title = "Distribution of Earthquake Magnitudes",  
    x = "Magnitude",  
    y = "Frequency"  
  ) +  
  theme_minimal()
```



```
df <- readr::read_delim(  
  "event_catalog_fixed.csv",  
  delim = ";",  
  col_types = cols(  
    Date = col_character(),  
    Longitude = col_double(),  
    Latitude = col_double(),  
    Depth = col_double(),  
    Rms = col_double(),  
    Type = col_character(),  
    Magnitude = col_double(),  
    Location = col_character(),  
    EventID = col_double()  
  )  
)
```

```
df <- df |>  
  mutate(  
    Date = parse_date_time(Date, orders = "d.m.Y H:M"),  
    Day = date(Date),  
    Hour = hour(Date)  
  )
```

```
file.exists("event_catalog_fixed.csv")
```

```
## [1] TRUE
```

```
head(df$Date, 10)
```

```
## [1] "2025-11-30 22:23:00 UTC" "2025-11-30 22:22:00 UTC"
## [3] "2025-11-30 22:17:00 UTC" "2025-11-30 22:16:00 UTC"
## [5] "2025-11-30 22:11:00 UTC" "2025-11-30 22:05:00 UTC"
## [7] "2025-11-30 21:55:00 UTC" "2025-11-30 21:53:00 UTC"
## [9] "2025-11-30 21:44:00 UTC" "2025-11-30 21:36:00 UTC"
```

```
head(df)
```

```
## # A tibble: 6 x 11
##   Date                Longitude Latitude Depth   Rms Type Magnitude Location
##   <dtm>              <dbl>    <dbl> <dbl> <dbl> <chr>    <dbl> <chr>
## 1 2025-11-30 22:23:00    33.3    39.1   7     0.51 ML      2.4 Kulu (Kony~
## 2 2025-11-30 22:22:00    36.7    38.2   7     0.82 ML      1.3 Göksun (Ka~
## 3 2025-11-30 22:17:00    38.2    38.3  11.6   0.35 ML      2.2 Yeşilyurt ~
## 4 2025-11-30 22:16:00    28.2    39.2  11.0   0.48 ML      1.2 Sındırgı (~
## 5 2025-11-30 22:11:00    26.0    39.1   6.69   0.62 ML      1.8 Ege Denizi~
## 6 2025-11-30 22:05:00    28.3    39.2   5.88   0.51 ML      1.2 Sındırgı (~
## # i 3 more variables: EventID <dbl>, Day <date>, Hour <int>
```

```
install.packages(c("tidyverse", "ggplot2", "lubridate", "readr", "dplyr"))
```

```
## Installing packages into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```
head(df$Date, 10)
```

```
## [1] "2025-11-30 22:23:00 UTC" "2025-11-30 22:22:00 UTC"
## [3] "2025-11-30 22:17:00 UTC" "2025-11-30 22:16:00 UTC"
## [5] "2025-11-30 22:11:00 UTC" "2025-11-30 22:05:00 UTC"
## [7] "2025-11-30 21:55:00 UTC" "2025-11-30 21:53:00 UTC"
## [9] "2025-11-30 21:44:00 UTC" "2025-11-30 21:36:00 UTC"
```

```
colnames(df)
```

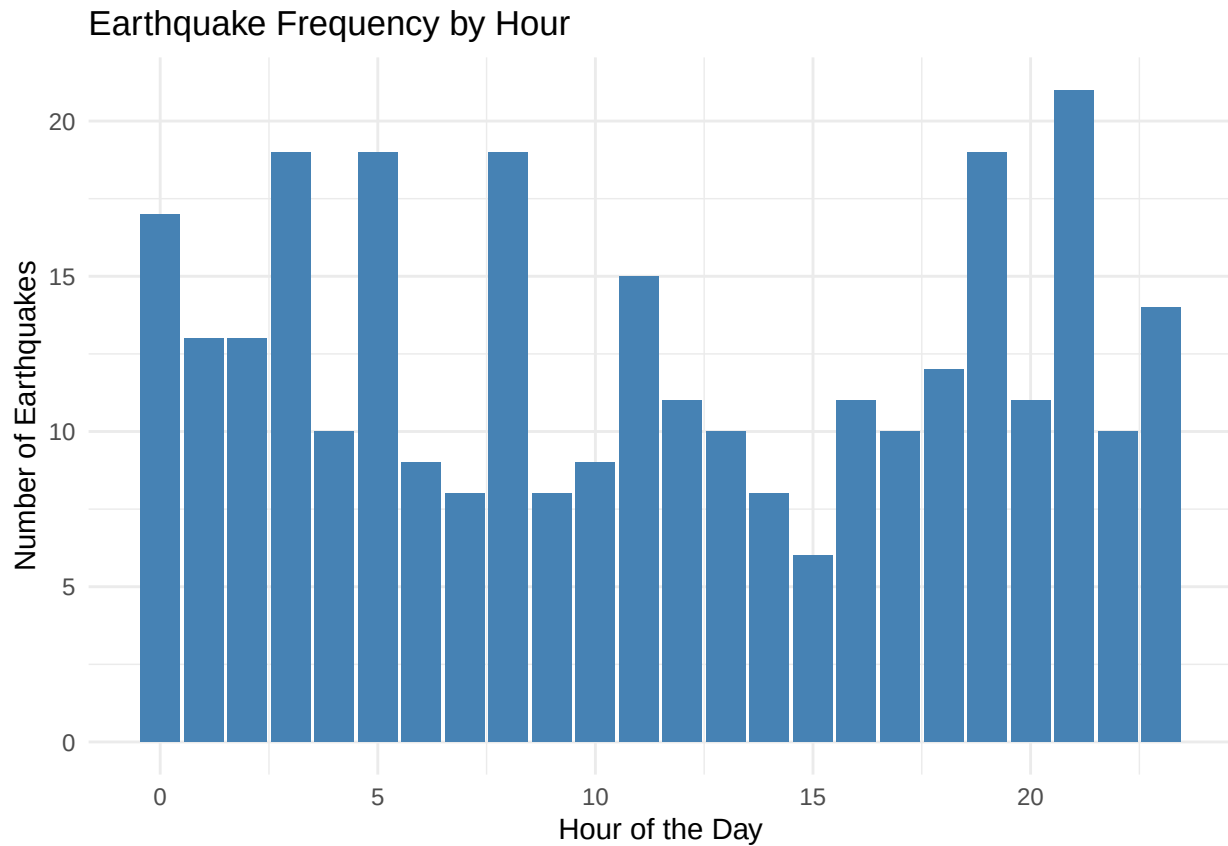
```
## [1] "Date"      "Longitude" "Latitude"  "Depth"    "Rms"      "Type"
## [7] "Magnitude" "Location"  "EventID"   "Day"      "Hour"
```

```
head(df[, c("Date", "Day", "Hour")], 10)
```

```
## # A tibble: 10 x 3
##   Date                Day        Hour
##   <dtm>              <date>    <int>
## 1 2025-11-30 22:23:00 2025-11-30    22
## 2 2025-11-30 22:22:00 2025-11-30    22
## 3 2025-11-30 22:17:00 2025-11-30    22
## 4 2025-11-30 22:16:00 2025-11-30    22
## 5 2025-11-30 22:11:00 2025-11-30    22
## 6 2025-11-30 22:05:00 2025-11-30    22
## 7 2025-11-30 21:55:00 2025-11-30    21
## 8 2025-11-30 21:53:00 2025-11-30    21
## 9 2025-11-30 21:44:00 2025-11-30    21
## 10 2025-11-30 21:36:00 2025-11-30    21
```

```
df %>%
  filter(!is.na(Hour)) %>%
  count(Hour) %>%
```

```
ggplot(aes(x = Hour, y = n)) +
  geom_col(fill = "steelblue") +
  labs(
    title = "Earthquake Frequency by Hour",
    x = "Hour of the Day",
    y = "Number of Earthquakes"
  ) +
  theme_minimal()
```



```
head(df$Date, 10)
```

```
## [1] "2025-11-30 22:23:00 UTC" "2025-11-30 22:22:00 UTC"
## [3] "2025-11-30 22:17:00 UTC" "2025-11-30 22:16:00 UTC"
## [5] "2025-11-30 22:11:00 UTC" "2025-11-30 22:05:00 UTC"
## [7] "2025-11-30 21:55:00 UTC" "2025-11-30 21:53:00 UTC"
## [9] "2025-11-30 21:44:00 UTC" "2025-11-30 21:36:00 UTC"
```

```
df <- df |>
  mutate(
    Date = parse_date_time(Date, orders = "dmy HM"),
    Day = date(Date),
    Hour = hour(Date)
  )
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `Date = parse_date_time(Date, orders = "dmy HM")`.
## Caused by warning:
## ! All formats failed to parse. No formats found.
```

```
df %>%
  filter(!is.na(Day)) %>%
  count(Day) %>%
  ggplot(aes(x = Day, y = n)) +
  geom_line(linewidth = 1.2, color = "tomato2") +
  geom_point(color = "black", size = 2) +
  labs(
    title = "Earthquake Count by Day",
    x = "Date",
    y = "Number of Earthquakes"
  ) +
  theme_minimal()
```

Earthquake Count by Day

Number of Earthquakes

Date

Findings 1. Distribution of Earthquake Magnitudes

The histogram shows that most earthquakes have magnitudes between 0.8 and 2.0. Higher magnitudes (above 3.0) are rare in this dataset. The distribution is right-skewed, meaning low-moderate magnitude earthquakes occur more frequently.

2. Earthquake Frequency by Hour

The bar plot indicates that earthquakes occur at all hours of the day, but some hours show slightly higher activity (e.g., 00:00–02:00, 05:00–07:00, 20:00–22:00). There is no perfectly regular pattern, but the activity is not evenly distributed.

3. Earthquake Count by Day

Earthquake counts increase significantly on November 29 compared to November 28, and remain high on November 30. This suggests a high-activity period within these days. Interpretation

The magnitude distribution suggests that the region experiences frequent but low-magnitude earthquakes. This is typical in areas with ongoing small-scale tectonic movement.

Hourly frequency variations may indicate natural seismic cycles or localized fault stress changes. However, there is no evidence of a strict time-based periodicity.

The sharp increase in total earthquakes around 29–30 November may indicate:

A temporary increase in stress on a local fault,

A swarm-like activity period,

Or aftershocks following a slightly stronger main event.